Chapter 4. Numerical Differentiation and Integration

4.1 Numerical Differentiation

Forward and Backward Differences

For small h,

$$f'(x) \approx \frac{f(x+h) - f(x_0)}{h}$$

$h > 0$: the forward-difference formula

$h < 0$: the backward-difference formula

Example 1

Use the forward-difference formula to approximate the derivative of $f(x) = \ln x$ at $x_0 = 1.8$ using $h = 0.1,\ h = 0.05$, and $h = 0.01$, and check the errors between the exact value $f'(1.8) = 5/9$.

```
f = @(x) log(x);
a = 1.8;
h_val = [0.1, 0.05, 0.01];
for h = h_val
    val = finite_difference(f,a,h);
    sprintf('the forward-difference: %.10f\t error: %.10f', val, abs(5/9 - val))
end
```

```
ans = 'the forward-difference: 0.5406722127      error: 0.0148833429'
ans = 'the forward-difference: 0.5479794838      error: 0.0075760718'
ans = 'the forward-difference: 0.5540180376      error: 0.0015375180'
```

Three-Point Midpoint Formula

$$f'(x_0) = \frac{1}{2h}[-f(x_0 - h) + f(x_0 + h)] - \frac{h^2}{6} f^{(3)}(\xi_1)$$

where ξ_1 lies between $x_0 - h$ and $x_0 + h$.

Example

Use the three-point midpoint formula to approximate $f'(0.9)$, where $f(x) = \sin x$ with $h = 0.1,\ 0.05,\ 0.02,\ 0.01,\ 0.005,\ 0.002,\ 0.001$, and check the errors between the exact value $f'(0.9) = \cos 0.9$.

```
f = @(x) sin(x);
a = 0.9;
h_val = [0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001];
exact_val = cos(a);
for h = h_val
    val = three_point_mid(f,a,h);
    sprintf('Three-Point Midpoint: %.10f\t error: %.10f', val, abs(val - cos(0.9)))
end
```

```
ans = 'Three-Point Midpoint: 0.6205744695        error: 0.0010354987'
ans = 'Three-Point Midpoint: 0.6213509965        error: 0.0002589718'
ans = 'Three-Point Midpoint: 0.6215685284        error: 0.0000414398'
ans = 'Three-Point Midpoint: 0.6215996082        error: 0.0000103601'
ans = 'Three-Point Midpoint: 0.6216073782        error: 0.0000025900'
ans = 'Three-Point Midpoint: 0.6216095539        error: 0.0000004144'
ans = 'Three-Point Midpoint: 0.6216098647        error: 0.0000001036'
```

5 decimal

```
f = @(x) sin(x);
a = 0.9;
h_val = [0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001];
exact_val = cos(a);
for h = h_val
    val = three_point_mid_5decimal(f,a,h);
    sprintf('Three-Point Midpoint: %.5f\t error: %.10f', val, abs(val - cos(0.9)));
end
```

4.4 Composite Numerical Integration

Composite Simpson's Rule

Theorem 4.4

Let $f \in C^4[a,b]$, n be even, $h = (b-a)/n$, and $x_j = a + jh$, for each $j = 0,\ 1,\ \cdots,\ n$. There exists a $\mu \in (a,b)$ for which the Composite Simpson's rule for n subintervals can be written with its error term as

$$\int_a^b f(x)\ dx = \frac{h}{3}\left[f(x_0) + 2\sum_{j=1}^{(n/2)-1} f(x_{2j}) + 4\sum_{j=1}^{n/2} f(x_{2j-1}) + f(x_n)\right] - \frac{(b-a)}{180} h^4 f^{(4)}(\mu).$$

Pseudo Code

INPUT endpoints a, b; even positive integer n.

OUTPUT approximation XI to I

Step 1 Set $h = (b-a)/n$.

Step 2 Set $XI0 = f(a) + f(b)$

$XI1$ = 0; (Summation of $f(x_{2i-1})$.)

$XI2$ = 0; (Summation of $f(x_{2i})$.)

Step 3 For $i = 1, 2, \cdots, n-1$ do Steps 4 and 5.

Step 4 Set $X = a + ih$.

Step 5 If i is even then set $XI2 = XI2 + f(X)$.

else set $XI1 = XI1 + f(X)$.

Step 6 Set $XI = h(XI0 + 2 \cdot XI2 + 4 \cdot XI1)/3$.

Step 7 OUTPUT (XI);

STOP.

Example

Approximate $\int_0^{\pi} \sin x \; dx$ with $n = 4, 8, 12, 16$, and 20.

```
f = @(x) sin(x);
nI = [4,8,12,16,20];
a = 0;
b = pi;
I = 2;
for n = nI
    XI = composite_simpson(f,a,b,n);
    sprintf("Composite Simpson's Rule: %.10f\t error: %.10f",XI, abs(XI - I))
end
```

```
ans = "Composite Simpson's Rule: 2.0045597550    error: 0.0045597550"
ans = "Composite Simpson's Rule: 2.0002691699    error: 0.0002691699"
ans = "Composite Simpson's Rule: 2.0000526243    error: 0.0000526243"
ans = "Composite Simpson's Rule: 2.0000165910    error: 0.0000165910"
ans = "Composite Simpson's Rule: 2.0000067844    error: 0.0000067844"
```

Composite Trapezoidal Rule

Theorem 4.5

Let $f \in C^2[a,b]$, $h = (b-a)/n$, and $x_j = a + jh$, for each $j = 0, 1, \cdots, n$. There exists a $\mu \in (a,b)$ for which the Composite Trapezoidal rule for n subintervals can be written with its error term as

$$\int_a^b f(x) \; dx = \frac{h}{2}\left[f(a) + 2\sum_{j=1}^{n-1} f(x_j) + f(b)\right] - \frac{(b-a)}{12} h^2 f''(\mu).$$

Example

Approximate $\int_0^{\pi} \sin x \; dx$ with $n = 4, 8, 12, 16$, and 20.

```
f = @(x) sin(x);
nI = [4,8,12,16,20];
a = 0;
b = pi;
I = 2;
for n = nI
    XI = composite_trapezoid(f,a,b,n);
    sprintf("Composite Trapezoidal Rule: %.10f\t error: %.10f",XI, abs(XI - I))
end
```

```
ans = "Composite Trapezoidal Rule: 1.8961188979  error: 0.1038811021"
ans = "Composite Trapezoidal Rule: 1.9742316019  error: 0.0257683981"
ans = "Composite Trapezoidal Rule: 1.9885637766  error: 0.0114362234"
ans = "Composite Trapezoidal Rule: 1.9935703438  error: 0.0064296562"
ans = "Composite Trapezoidal Rule: 1.9958859727  error: 0.0041140273"
```

Composite Midpoint Rule

Theorem 4.6

Let $f \in C^2[a,b]$, n be even, $h = (b-a)/(n+2)$, and $x_j = a + (j+1)h$, for each $j = -1, 0, \cdots, n+1$. There exists a $\mu \in (a,b)$ for which the Composite Midpoint rule for $n+2$ subintervals can be written with its error term as

$$\int_a^b f(x) \; dx = 2h\sum_{j=0}^{n/2} f(x_{2j}) + \frac{(b-a)}{6} h^2 f''(\mu).$$

Example

Approximate $\int_0^{\pi} \sin x \ dx$ with $n = 4,\ 8,\ 12,\ 16,$ and 20.

```
f = @(x) sin(x);
nI = [4,8,12,16,20];
a = 0;
b = pi;
I = 2;
for n = nI
    XI = composite_midpoint(f,a,b,n);
    sprintf("Composite Midpoint Rule: %.10f\t error: %.10f",XI, abs(XI - I))
end
```

```
ans = "Composite Midpoint Rule: 1.8137993642     error: 0.1862006358"
ans = "Composite Midpoint Rule: 1.9337655981     error: 0.0662344019"
ans = "Composite Midpoint Rule: 1.9663166788     error: 0.0336833212"
ans = "Composite Midpoint Rule: 1.9796508112     error: 0.0203491888"
ans = "Composite Midpoint Rule: 1.9863869866     error: 0.0136130134"
```

4.7 Gaussian Quadrature

n	Roots $r_{n,i}$	Coefficients $c_{n,i}$
2	0.5773502692	1.0000000000
	−0.5773502692	1.0000000000
3	0.7745966692	0.5555555556
	0.0000000000	0.8888888889
	−0.7745966692	0.5555555556
4	0.8611363116	0.3478548451
	0.3399810436	0.6521451549
	−0.3399810436	0.6521451549
	−0.8611363116	0.3478548451
5	0.9061798459	0.2369268850
	0.5384693101	0.4786286705
	0.0000000000	0.5688888889
	−0.5384693101	0.4786286705
	−0.9061798459	0.2369268850

An integral $\int_a^b f(x) \ dx$ over an arbitrary $[a, b]$ can be transformed into an integral over $[-1, 1]$ by using the change of variables:

$$t = \frac{2x - a - b}{b - a} \quad \Longleftrightarrow \quad x = \frac{1}{2}[(b - a)t + a + b].$$

This permits Gaussian quadrature to be applied to any interval $[a, b]$, because

$$\int_a^b f(x) \ dx = \int_{-1}^{1} f\left(\frac{(b - a)t + (b + a)}{2}\right) \frac{(b - a)}{2} \ dt.$$

Example 2

Consider the integral $\int_1^3 x^6 - x^2 \sin(2x) \ dx = 317.3442466.$

```
f = @(x) x^6 - x^2 * sin(2*x);
val = gaussian_quadrature(1,3,2,f);
disp(val)
```

```
306.8199
```